

→ head (iris_data)

	sepal.length	sepal.width	petal.length	petal.width
1.	5.1	3.5	1.4	0.2
2.	4.9	3.0	1.4	0.2
3.	4.7	3.2	1.3	0.2
4.	4.6	3.1	1.5	0.2
5.	5.0	3.6	1.4	0.2
6.	5.4	3.9	1.7	0.4

Ex: No: 8
Date: 22/4/20

Iris Flower Clustering Using K-Means Algorithm

Problem Description:

The Iris dataset contains 150 samples of three iris species (Setosa, Versicolour, Virginica) with four numerical features: Sepal Length, Sepal Width, Petal Length, Petal Width. The goal is to group the flowers into clusters based on these features using the K-Means clustering algorithm, without using the species labels.

Objective:

The objective of this program is to implement the K-Means clustering algorithm in R to group iris flowers based on their numerical attributes, visualize the clusters, and evaluate clustering performance against the actual species.

Procedure:

- Step 1: Install and Load Required Libraries
We install and load the necessary libraries for implementing K-Means clustering.

```
# Install and Load required libraries
install.packages("factextra")
library(ggplot2)
library(cluster)
library(factextra)
```

- Step 2: Load and Prepare the Dataset

We use the iris dataset (excluding the Species column) for clustering.

```
# Load the iris dataset
data <- iris

# Remove the Species column (only numeric features)
iris_data <- iris[, -5]

# Display first few rows
head(iris_data)
```

total width sum of square



[Handwritten scribbles]

22

[145] 3 3 2 3 2

cluster

56 0 0 0

3
8
3

2 62 ~~5 42~~

~~0.45~~

~~Shoppers~~ ~~at~~ ~~hood~~

- ```
Elbow method to find optimal k using fviz_nbclust()
```

**Interpretation:**

The optimal  $k$  is chosen where the within-cluster sum of squares (WSS) starts to level off (elbow point).

- ### # Apply K-Means clustering with 3 clusters

# Print cluster centers

Prins (k means model & centers)

```
Print cluster assignments
```

Prim (kmeans-model of cluster)

- **Step 5: Visualize Clusters**  
We plot the clusters using Principal Component Analysis (PCA) for dimensionality reduction.

```
Visualize clusters - fviz_cluster()
```

5. V1 Z-clusters (k means model, data = iris-data, guess = "police", all pr. type = "lonkers") +

Agitate (k-Means clustering on this Dataset")

k-Means clustering on Iris Dataset.

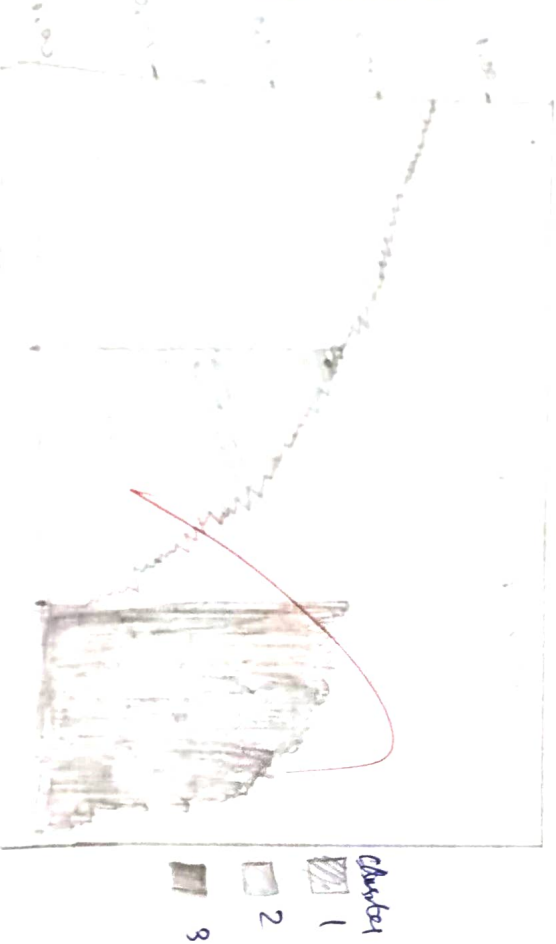


Dim 1 (73%)

Cluster 1  
Cluster 2  
Cluster 3

clusters silhouette plot

Average silhouette width: 0.55



Silhouette width si

- Step 6: Evaluate Cluster Performance  
We calculate the Silhouette Score to measure clustering quality

# Compute silhouette score

$\text{silhouette\_score} \leftarrow \text{silhouette}(kmeans\_model \&$   
 $\text{cluster, dist(iris\_data)})$

# Plot silhouette analysis

$\text{fviz\_silhouette}(\text{silhouette\_score})$

Completed  
24/04/2026

Result:

Thus, the K-Means clustering algorithm was successfully implemented in R, and the resulting clusters were visualized and evaluated, effectively revealing the natural groupings in the Iris dataset.